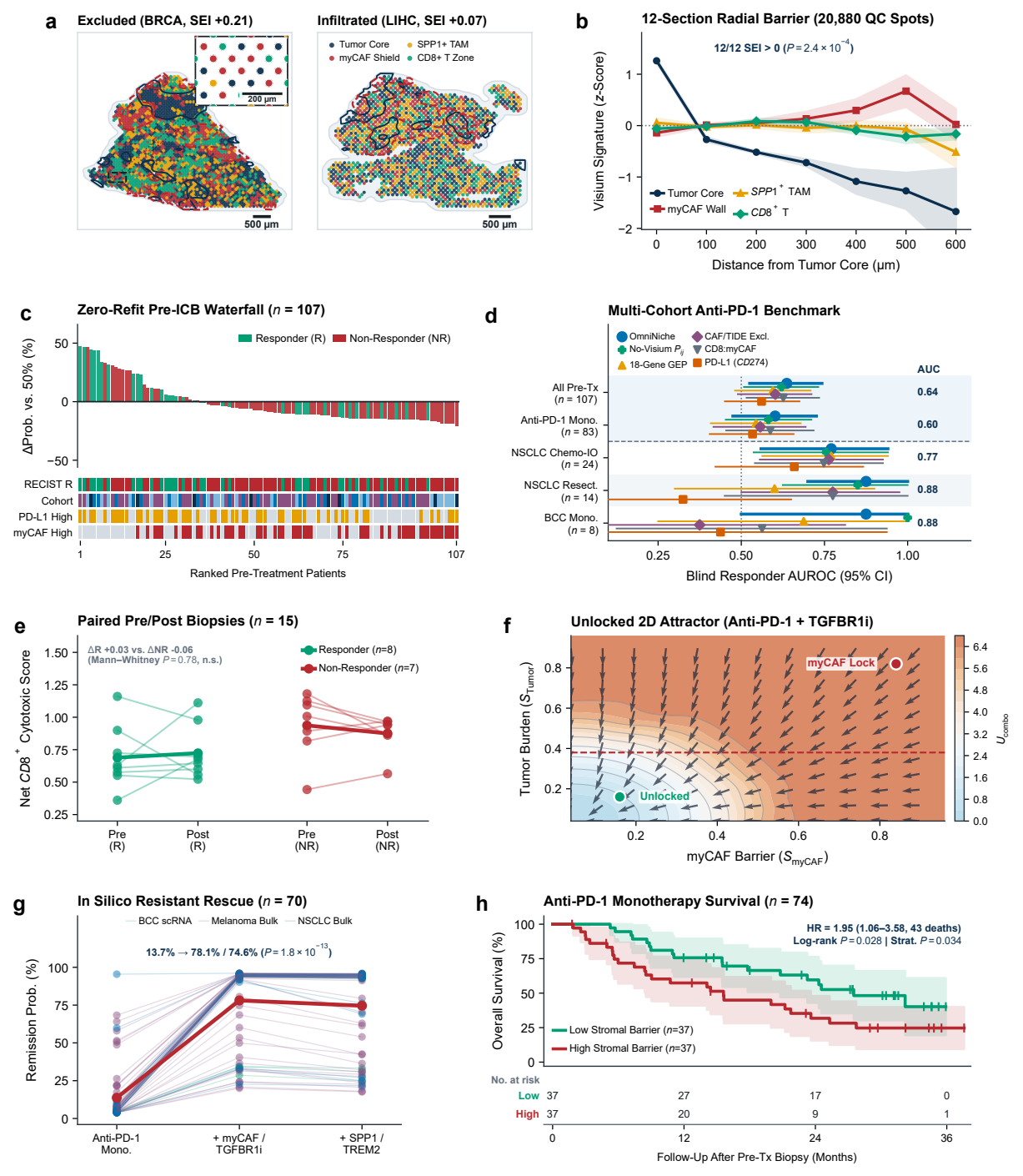

a
Excluded (BRCA, SEI +0.21)
200 µm
500 µm
Infiltrated (LIHC, SEI +0.07)
Tumor Core
SPP1+ TAM
myCAF Shield
CD8+ T Zone
500 µm
b
12-Section Radial Barrier (20,880 QC Spots)
12/12 SEI > 0 (P = 2.4 × 10−4)
Visium Signature (z-Score)
Tumor Core
SPP1+ TAM
myCAF Wall
CD8+ T
Distance from Tumor Core (µm)
c
Zero-Refit Pre-ICB Waterfall (n = 107)
Responder (R)
Non-Responder (NR)
ΔProb. vs. 50% (%)
RECIST R
Cohort
PD-L1 High
myCAF High
Ranked Pre-Treatment Patients
d
Multi-Cohort Anti-PD-1 Benchmark
OmniNiche
CAF/TIDE Excl.
No-Visium Pij
CD8:myCAF
18-Gene GEP
PD-L1 (CD274)
AUC
All Pre-Tx (n = 107)
Anti-PD-1 Mono. (n = 83)
NSCLC Chemo-IO (n = 24)
NSCLC Resect. (n = 14)
BCC Mono. (n = 8)
0.64
0.60
0.77
0.88
0.88
Blind Responder AUROC (95% CI)
e
Paired Pre/Post Biopsies (n = 15)
ΔR +0.03 vs. ΔNR -0.06
(Mann–Whitney P = 0.78, n.s.)
Responder (n=8)
Non-Responder (n=7)
Net CD8+ Cytotoxic Score
Pre (R)
Post (R)
Pre (NR)
Post (NR)
f
Unlocked 2D Attractor (Anti-PD-1 + TGFBR1i)
myCAF Lock
Unlocked
Tumor Burden (STumor)
myCAF Barrier (SmyCAF)
Ucombo
g
In Silico Resistant Rescue (n = 70)
BCC scRNA
Melanoma Bulk
NSCLC Bulk
13.7% → 78.1% / 74.6% (P = 1.8 × 10−13)
Remission Prob. (%)
Anti-PD-1 Mono.
+ myCAF / TGFBR1i
+ SPP1 / TREM2
h
Anti-PD-1 Monotherapy Survival (n = 74)
HR = 1.95 (1.06–3.58, 43 deaths)
Log-rank P = 0.028 | Strat. P = 0.034
Overall Survival (%)
Low Stromal Barrier (n=37)
High Stromal Barrier (n=37)
No. at risk
Low 37 27 17 0
High 37 20 9 1
Follow-Up After Pre-Tx Biopsy (Months)